Certificate

This is to certify that the mini project titled **"Creating and Extracting Archives Using Linux Commands"** presents a practical study of Linux file archiving and compression techniques and demonstrates the use of command-line utilities to create, compress, extract, and verify file archives in a CentOS environment.

The project report covers the following key areas:

- Introduction to Linux file archiving and compression
- Creation of directories and sample files using Linux commands
- Creating TAR archives using the tar command
- Compressing archives using gzip
- Extracting compressed archives into a separate directory
- Testing and verification of extracted files using Linux commands
- Results, limitations, future enhancements, and conclusion

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I would like to express my sincere gratitude to my project guide and faculty members for their guidance, suggestions, and academic support during the preparation of the **Creating and Extracting Archives Using Linux Commands** mini project. Their feedback helped me understand Linux file management, archive creation, compression, extraction, and verification using command-line utilities.

I also appreciate my classmates, friends, and family for their encouragement and support throughout the completion of this project. Finally, I am thankful to Chandigarh University for providing an academic environment in which I could develop my practical knowledge of Linux commands, TAR archives, GZIP compression, and file extraction techniques.

Abstract

The **Creating and Extracting Archives Using Linux Commands** mini project is a practical Linux system administration project carried out in a CentOS virtual machine using Oracle VirtualBox. It focuses on file archiving, compression, extraction, and verification using command-line utilities such as tar and gzip.
The project demonstrates the creation of directories and sample files, combining multiple files into a single TAR archive, compressing the archive into a .tar.gz file, and extracting the compressed archive into a separate directory. The extracted files are then verified against the original files to ensure that their contents remain unchanged. The report presents the introduction, objectives, scope, software requirements, commands used, implementation steps, testing and verification, results, limitations, future enhancements, and conclusion. The project is intended for academic demonstration and provides a foundation for understanding Linux file management and data archiving techniques.

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Chapter 1 – Introduction

1.1 Background

File archiving and compression are important tasks in Linux system administration. They help users combine multiple files into a single archive, reduce storage requirements, and transfer or store data conveniently. Linux provides command-line utilities such as tar and gzip to perform these operations efficiently.

The tar command is used to combine files and directories into a single archive, while gzip compresses files to reduce their size. Compressed archives can later be extracted to restore the original files. These operations are useful for organising files, transferring data, and preparing backups.

This project demonstrates archive creation, compression, extraction, and verification using CentOS running in Oracle VirtualBox.

1.2 Objectives

- Understand the basics of Linux file archiving and compression.
- Create directories and sample files using Linux commands.
- Combine multiple files into a single TAR archive.
- Compress archives using the GZIP utility.
- Extract compressed archives into a separate directory.
- Verify extracted files against the original files.
- Develop practical skills in Linux command-line operations.

1.3 Scope

The project covers the creation and extraction of file archives in a CentOS virtual machine. It demonstrates the use of tar and gzip to create .tar and .tar.gz files, extract archived content, and verify the integrity of restored files.

The project is intended for academic learning and basic file management. Advanced features such as scheduled backups, automated Bash scripts, remote backups, and backup recovery systems are outside the current scope and may be considered for future enhancement.

CHAPTER 2 – TOOLS AND TECHNOLOGIES USED

2.1 CentOS and Oracle VirtualBox

CentOS is a Linux distribution that provides a command-line environment for learning common system-administration tasks. In this project, CentOS is used to execute commands for directory creation, file creation, archive creation, compression, extraction, and verification.

Oracle VirtualBox is virtualization software that allows an operating system to run inside a virtual machine on a host computer. Using a virtual machine provides a contained environment for practical exercises. The sample data for this project is created inside the CentOS user's home directory.

2.2 TAR Utility

The tar utility packages files and directories into one archive. The archive can preserve directory structure and relevant file metadata. TAR itself is an archiving tool; it does not necessarily compress the data.

Command:

tar -cvf myfiles.tar myfiles/

The command contains the following options:

- tar: Invokes the TAR utility.
- -c: Creates a new archive.
- -v: Displays the files being processed.
- -f: Specifies the archive filename.
- myfiles.tar: The output archive filename.
- myfiles/: The directory being archived.

2.3 GZIP Utility

The gzip utility compresses a file. When GZIP is run on a TAR archive, it normally replaces the original .tar file with a compressed .tar.gz file.

Command:

gzip myfiles.tar

After successful compression, the compressed file is named myfiles.tar.gz. The original source directory remains unchanged.

2.4 TAR and GZIP Together

TAR and GZIP can be used together to create a compressed archive in one step.

Command:

tar -czvf final_backup.tar.gz myfiles/

The -z option tells TAR to use GZIP compression while creating the archive. This is a convenient way to combine archiving and compression into a single operation.

The resulting .tar.gz file contains the archived directory and its files in compressed form.

CHAPTER 3 – SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The project has modest hardware requirements because it uses only a small number of text files and standard command-line utilities.

The required hardware includes:

- A computer capable of running Oracle VirtualBox.
- Sufficient RAM to run the existing CentOS virtual machine.
- Available disk space for the virtual machine and project files.
- A keyboard and display for interacting with the terminal.

3.2 Software Requirements

The software requirements are:

- Oracle VirtualBox installed on the host computer.
- A working CentOS virtual machine.
- A terminal or shell session in CentOS.
- The tar utility.
- The gzip utility.
- Standard commands such as mkdir, echo, ls, cat, and diff.

3.3 Initial Checks

Before starting the implementation, the Linux terminal was opened and the availability of TAR and GZIP was checked.

Command to check TAR:

tar --version

Command to check GZIP:

gzip --version

These commands display version information when the utilities are available. If a utility is missing, it may need to be installed using the package manager configured for that CentOS system.

3.4 Project Directory Structure

The project is organised under a directory named LinuxMiniProject in the user's home directory.

The structure is:

LinuxMiniProject/

- myfiles/
  - file1.txt
  - file2.txt
  - file3.txt
- myfiles.tar.gz
- final_backup.tar.gz
- extracted_files/
  - myfiles/
    - file1.txt
    - file2.txt
    - file3.txt

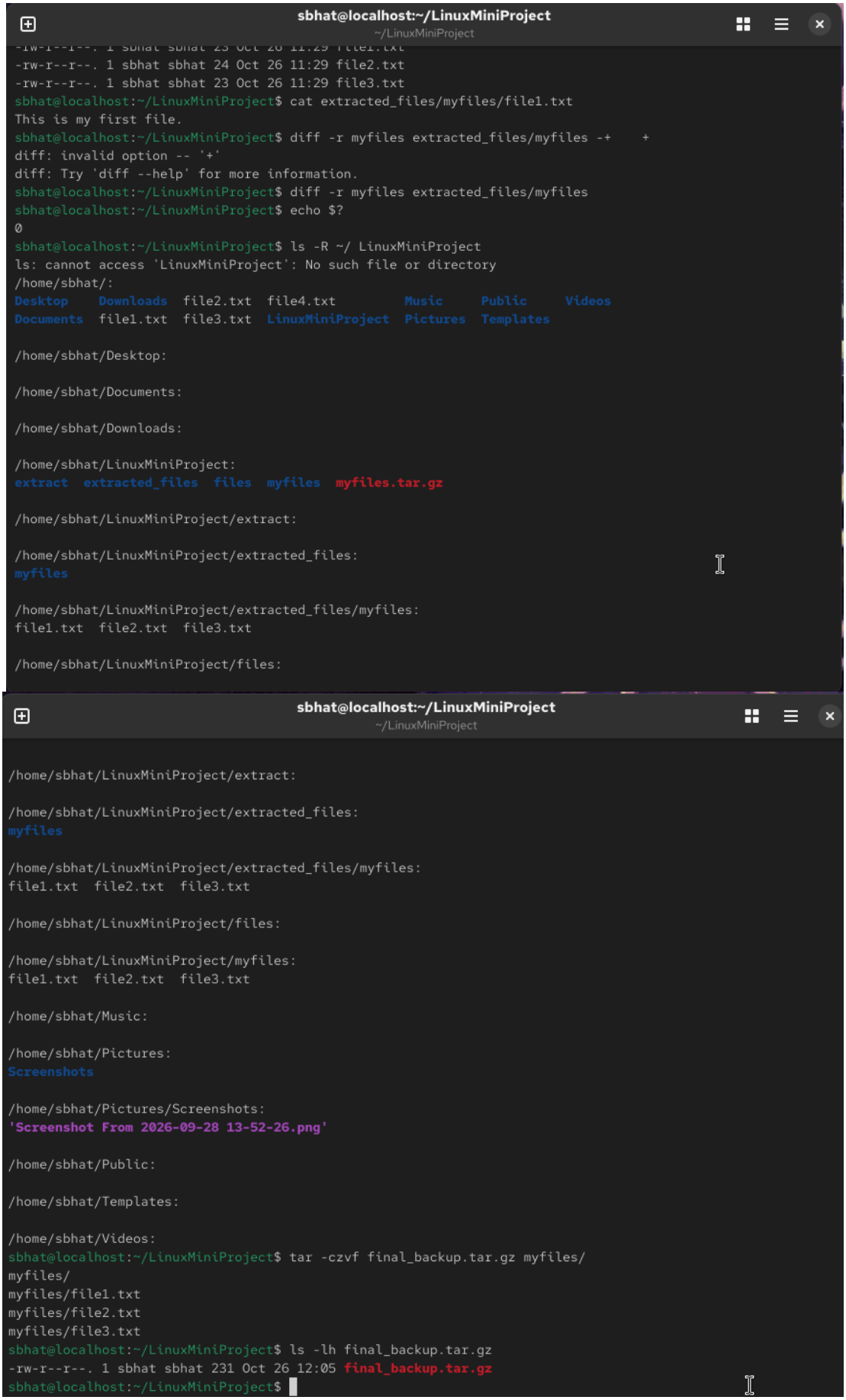
The myfiles directory contains the original sample files. The compressed archive is stored in the project directory, while extracted_files is used as the destination for restored content.

CHAPTER 4 – IMPLEMENTATION

```
sbhat@localhost:~/LinuxMiniProject$ mkdir myfiles
sbhat@localhost:~/LinuxMiniProject$ echo "This is my first file." > myfiles/file1.txt
sbhat@localhost:~/LinuxMiniProject$ echo "This is my second file." > myfiles/file2.txt
sbhat@localhost:~/LinuxMiniProject$ echo "This is my third file." > myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ ls -l myfiles
total 12
-rw-r--r--. 1 sbhat sbhat 23 Oct 26 11:29 file1.txt
-rw-r--r--. 1 sbhat sbhat 24 Oct 26 11:29 file2.txt
-rw-r--r--. 1 sbhat sbhat 23 Oct 26 11:29 file3.txt
sbhat@localhost:~/LinuxMiniProject$ cat myfiles/file1.txt
This is my first file.
sbhat@localhost:~/LinuxMiniProject$ tar -cvf myfiles.tar /myfiles
tar: Removing leading `/' from member names
tar: /myfiles: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
sbhat@localhost:~/LinuxMiniProject$ tar -cvf myfiles.tar myfiles/
myfiles/
myfiles/file1.txt
myfiles/file2.txt
myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ ls -lh myfiles.tar
-rw-r--r--. 1 sbhat sbhat 10K Oct 26 11:36 myfiles.tar
sbhat@localhost:~/LinuxMiniProject$ tar -tvf myfiles.tar
drwxr-xr-x sbhat/sbhat       0 2026-10-26 11:29 myfiles/
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file1.txt
-rw-r--r-- sbhat/sbhat      24 2026-10-26 11:29 myfiles/file2.txt
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$
```

```
sbhat@localhost:~/LinuxMiniProject$ tar -tvf myfiles.tar
drwxr-xr-x sbhat/sbhat       0 2026-10-26 11:29 myfiles/
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file1.txt
-rw-r--r-- sbhat/sbhat      24 2026-10-26 11:29 myfiles/file2.txt
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ gzip myfiles.tar
sbhat@localhost:~/LinuxMiniProject$ ls -lh myfiles.tar.gz
-rw-r--r--. 1 sbhat sbhat 243 Oct 26 11:36 myfiles.tar.gz
sbhat@localhost:~/LinuxMiniProject$ tar -tvf myfiles.tar.gz
drwxr-xr-x sbhat/sbhat       0 2026-10-26 11:29 myfiles/
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file1.txt
-rw-r--r-- sbhat/sbhat      24 2026-10-26 11:29 myfiles/file2.txt
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$
```

```
sbhat@localhost:~/LinuxMiniProject$ tar -tvf myfiles.tar.gz
drwxr-xr-x sbhat/sbhat       0 2026-10-26 11:29 myfiles/
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file1.txt
-rw-r--r-- sbhat/sbhat      24 2026-10-26 11:29 myfiles/file2.txt
-rw-r--r-- sbhat/sbhat      23 2026-10-26 11:29 myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ mkdir extract files
sbhat@localhost:~/LinuxMiniProject$ tar -xzvf myfiles.tar.gz -C extract files/
tar: files: Not found in archive
tar: Exiting with failure status due to previous errors
sbhat@localhost:~/LinuxMiniProject$ tar -xzvf myfiles.tar.gz -C extractfiles/
tar: extractfiles: Cannot open: No such file or directory
tar: Error is not recoverable: exiting now
sbhat@localhost:~/LinuxMiniProject$ mkdir extracted_files
sbhat@localhost:~/LinuxMiniProject$ tar -xzvf myfiles.tar.gz -C extracted_files/
myfiles/
myfiles/file1.txt
myfiles/file2.txt
myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ ls -l extracted_files/myfiles/
total 12
-rw-r--r--. 1 sbhat sbhat 23 Oct 26 11:29 file1.txt
-rw-r--r--. 1 sbhat sbhat 24 Oct 26 11:29 file2.txt
-rw-r--r--. 1 sbhat sbhat 23 Oct 26 11:29 file3.txt
sbhat@localhost:~/LinuxMiniProject$ cat extracted_files/myfiles/file1.txt
This is my first file.
sbhat@localhost:~/LinuxMiniProject$
```


sbhat@localhost:~/LinuxMiniProject
~/LinuxMiniProject
-rw-r--r--. 1 sbhat sbhat 24 Oct 26 11:29 file2.txt
-rw-r--r--. 1 sbhat sbhat 23 Oct 26 11:29 file3.txt
sbhat@localhost:~/LinuxMiniProject$ cat extracted_files/myfiles/file1.txt
This is my first file.
sbhat@localhost:~/LinuxMiniProject$ diff -r myfiles extracted_files/myfiles -+ +
diff: invalid option -- '+'
diff: Try 'diff --help' for more information.
sbhat@localhost:~/LinuxMiniProject$ diff -r myfiles extracted_files/myfiles
sbhat@localhost:~/LinuxMiniProject$ echo $?
0
sbhat@localhost:~/LinuxMiniProject$ ls -R ~/ LinuxMiniProject
ls: cannot access 'LinuxMiniProject': No such file or directory
/home/sbhat/:
Desktop Downloads file2.txt file4.txt Music Public Videos
Documents file1.txt file3.txt LinuxMiniProject Pictures Templates
/home/sbhat/Desktop:
/home/sbhat/Documents:
/home/sbhat/Downloads:
/home/sbhat/LinuxMiniProject:
extract extracted_files files myfiles myfiles.tar.gz
/home/sbhat/LinuxMiniProject/extract:
/home/sbhat/LinuxMiniProject/extracted_files:
myfiles
/home/sbhat/LinuxMiniProject/extracted_files/myfiles:
file1.txt file2.txt file3.txt
/home/sbhat/LinuxMiniProject/files:
sbhat@localhost:~/LinuxMiniProject
~/LinuxMiniProject
/home/sbhat/LinuxMiniProject/extract:
/home/sbhat/LinuxMiniProject/extracted_files:
myfiles
/home/sbhat/LinuxMiniProject/extracted_files/myfiles:
file1.txt file2.txt file3.txt
/home/sbhat/LinuxMiniProject/files:
/home/sbhat/LinuxMiniProject/myfiles:
file1.txt file2.txt file3.txt
/home/sbhat/Music:
/home/sbhat/Pictures:
Screenshots
/home/sbhat/Pictures/Screenshots:
'Screenshot From 2026-09-28 13-52-26.png'
/home/sbhat/Public:
/home/sbhat/Templates:
/home/sbhat/Videos:
sbhat@localhost:~/LinuxMiniProject$ tar -czvf final_backup.tar.gz myfiles/
myfiles/
myfiles/file1.txt
myfiles/file2.txt
myfiles/file3.txt
sbhat@localhost:~/LinuxMiniProject$ ls -lh final_backup.tar.gz
-rw-r--r--. 1 sbhat sbhat 231 Oct 26 12:05 final_backup.tar.gz
sbhat@localhost:~/LinuxMiniProject$

CHAPTER 7 – CONCLUSION

The mini project “Creating and Extracting Archives Using Linux Commands” demonstrates the basic process of archiving and compressing files in a CentOS environment running in Oracle VirtualBox.

The project begins by creating a working directory and three sample text files. The tar utility is used to combine the files into an archive, and gzip is then used to compress that archive.

The compressed archive is extracted into a separate directory using tar with the extraction and GZIP options. The restored files are inspected with ls and cat, and the diff command is used to compare the original and extracted directories. An exit status of 0 from diff indicates that no differences were detected.

The practical work helped develop familiarity with Linux directory navigation, file creation, command options, archive inspection, compression, extraction, and basic verification. It also demonstrated the importance of checking results at each stage instead of assuming that a command succeeded.

The project is limited to a small local demonstration and does not implement scheduled backups, encryption, remote storage, or a complete recovery strategy. Nevertheless, it provides a useful foundation for future work involving Bash automation and backup management.

In conclusion, TAR and GZIP are useful Linux utilities for packaging and compressing files. By understanding their commands and options, users can manage collections of files more efficiently and restore archived data when needed.